Andy Baek

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Education

Boston University **Sep 2024 – Expected May 2028**
Bachelor of Science in Data Science *Boston, Massachusetts*
GPA: 3.8/4.0 (Dean's List)

Experience

BU Psychology Lab **Sept 2026 – Present**
Software Engineer | React, React Native, FastAPI, Supabase *Boston, MA*

- Building a full-stack experimental platform with a 3-person team for a media psychology lab studying short-form video engagement, pairing a researcher web app with an Android tablet app that mimics Reels/Shorts-style interfaces.
- Developing millisecond-precision behavioral event capture (watch time, swipes, button presses, survey first-touch) with video pre-loading to keep data valid for human-subject research.
- Implementing block-based experiment configuration with subset randomization, cloud sync, and structured CSV/ZIP export for researcher analysis.

Boston University **Sept 2026 – Present**
Course Assistant, DS120 (Foundations of Data Science) *Boston, MA*

- Hold weekly office hours supporting students through proof-based coursework spanning mathematical logic, linear algebra, multivariable calculus, and probability theory.
- Grade weekly problem sets and exams for a 180+ student course, applying rubrics consistently across topics including linear systems, gradient descent, and hypothesis testing.

Hyundai Motor Manufacturing Alabama, LLC **June 2024 – Aug 2024**
Inventory Auditor *Montgomery, AL*

- Performed plant-wide inventory verification on a 1,500 vehicles/day assembly line, serving as the final check after section-level inspections and enabling same-shift restocking.
- Coordinated with materials and production teams to resolve inventory discrepancies in real time, preventing line stoppages during high-volume shifts.

Projects

Tennis Player & Ball Tracker | *Python, YOLOv8, OpenCV* **Sept 2025**

- Built an end-to-end pipeline that ingests full-length match footage, automatically segments it into live-play rallies, and outputs a condensed highlight video with tracking stats overlaidreducing footage by 70%.
- Fine-tuned YOLOv8 on tennis match footage, boosting ball detection from near-zero (pretrained baseline) to **80–90%** of frames; closed remaining gaps with trajectory interpolation to produce continuous ball tracks.
- Trained a court keypoint detection model and computed a **homography** to project player and ball positions onto a top-down 2D mini-court, enabling automated shot and rally analysis.

Flappy Bird RL – Tabular Q-Learning vs. DQN with Behavioral Cloning | *Python, Gymnasium, PyTorch* **Mar 2026**

- Built a custom Flappy Bird environment (Pygame + Gymnasium) and implemented tabular Q-learning and DQN agents in PyTorch to compare classical vs. deep RL on the same task.
- Bootstrapped both agents with behavioral cloning on expert demonstrations before RL fine-tuning, overcoming slow cold-start learning from sparse rewards.
- Diagnosed a sparse-reward failure where agents rarely cleared the first pipe; added a gap-alignment reward-shaping term that took the final agent from failing at the first obstacle to surviving indefinitely (1,000+ pipes without failure).

Fair Best-Arm Identification – Sequential Decision Making | *Python* **April 2026**

- Implemented the F-TaS (Fair Track-and-Stop) algorithm with a partner from a 2024 fairness-constrained bandit paper and reproduced its experiments, benchmarking against TaS and a uniform-sampling baseline across 4 fairness conditions.
- Ran 100-trial experiments ($K = 10$ arms, $\delta = 0.1$) showing F-TaS was the only algorithm to keep fairness violations near zero while matching TaS's sample efficiency; unconstrained TaS violated fairness by up to 18–20% in adversarial settings.
- Designed a novel experiment beyond the original paper: swept δ from 10^{-1} to 10^{-4} and fit a power law to fairness violations, finding a shallow decay ($\approx \delta^{0.03}$) that reveals a practical gap in the algorithm's asymptotic guarantee.

Relevant Coursework

AI/ML: Neural Networks, Reinforcement Learning, Sequential Decision Making (Bandits, Concentration Inequalities), Probabilistic ML
CS & Math: Data Structures, Algorithms, Database Design, Cryptography, Linear Algebra, Probability & Statistics

Technical Skills

Languages: Python, Rust, JavaScript/TypeScript, R, OCaml, HTML/CSS, Shell/Bash
Developer Tools: Git/GitHub, VS Code, Jupyter
Databases: PostgreSQL (Supabase), MongoDB, SQL
Libraries/Frameworks: PyTorch, YOLOv8, OpenCV, React, React Native, FastAPI, MediaPipe, scikit-learn, Seaborn, Matplotlib, Beautiful Soup